

GitHub AWS OIDC Authentication

GitHub Actions can securely authenticate with AWS using OIDC (OpenID Connect) without storing long-term AWS access keys in GitHub secrets. This is considered a modern and secure authentication method for CI/CD pipelines.

What is OIDC?

OIDC (OpenID Connect) allows GitHub Actions to request temporary AWS credentials dynamically during workflow execution. Instead of storing AWS Access Key and Secret Key in GitHub, AWS trusts GitHub's identity provider and issues short-lived credentials.

Benefits of GitHub OIDC Authentication

- No hardcoded AWS credentials
- Improved security posture
- Temporary short-lived credentials
- Easier credential rotation management
- Reduced risk of secret leakage

High-Level Flow

1. GitHub Actions workflow requests OIDC token.
2. GitHub sends identity token to AWS.
3. AWS IAM validates trust relationship.
4. AWS STS issues temporary credentials.
5. Workflow uses temporary credentials for deployment.

AWS Configuration Steps

1. Create OIDC Identity Provider in AWS IAM.
2. Configure GitHub as trusted provider.
3. Create IAM Role with required permissions.
4. Add trust policy for GitHub repository.
5. Configure GitHub Actions workflow.

Sample Trust Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Federated": "arn:aws:iam::<AWS_ACCOUNT_ID>:oidc-provider/token.actions.githubusercontent.com"
      },
      "Action": "sts:AssumeRoleWithWebIdentity",
      "Condition": {
        "StringEquals": {
          "token.actions.githubusercontent.com:aud": "sts.amazonaws.com"
        }
      }
    }
  ]
}
```

```

    },
    "StringLike": {
      "token.actions.githubusercontent.com:sub": "repo:username/repository:*"
    }
  }
}
]
}

```

GitHub Actions Workflow Example

```

name: Deploy to AWS

on:
  push:
    branches:
      - main

permissions:
  id-token: write
  contents: read

jobs:
  deploy:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Configure AWS Credentials
        uses: aws-actions/configure-aws-credentials@v4
        with:
          role-to-assume: arn:aws:iam::<ACCOUNT_ID>:role/github-oidc-role
          aws-region: ap-south-1

      - name: Verify Identity
        run: aws sts get-caller-identity

```

Common Issues

- Incorrect trust policy conditions
- Missing id-token permission in workflow
- Wrong repository name in trust relationship
- IAM role missing required permissions
- OIDC provider not configured correctly

Best Practices

- Use least privilege IAM policies
- Restrict repository access using StringLike conditions
- Avoid wildcard permissions
- Use environment-specific IAM roles
- Monitor CloudTrail logs for AssumeRole activity

GitHub OIDC authentication is now considered the preferred and secure method for AWS authentication in CI/CD pipelines. It eliminates the need for static AWS credentials and improves overall DevSecOps posture.